# CURRICULUM VITAE

## PERSONAL INFORMATION




**NAME:** Wookyoung Kim
**DATE OF BIRTH:** September 27, 1993
**NATIONALITY:** Korea
**AFFILIATION:** Korea Institute of Machinery and Materials (KIMM), Heat Pump Research Center
**POSITION:** Associate Professor, University of Science and Technology (UST) (2026-present)
Senior Researcher, Korea Institute of Machinery and Materials (KIMM) (2021-present)
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## EDUCATION

| Degree | University | Department | Date |
|---|---|---|---|
| Ph.D. | KAIST | Department of Mechanical Engineering | February, 2021 |
| | **Dissertation:** *"Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes"* | | |
| | **Advisor:** Sung Jin Kim | | |
| M.S. | KAIST | Department of Mechanical Engineering | February, 2017 |
| | **Thesis:** *"Effect of artificial cavities on the thermal performance of pulsating heat pipes"* | | |
| | **Advisor:** Sung Jin Kim | | |
| B.S. | KAIST | Department of Mechanical Engineering | February, 2015 |

## RESEARCH INTERESTS

(1) Data Center Cooling (Immersion cooling, Direct liquid cooling, Jet impingement, PUE optimization)
(2) High-Heat-Flux Electronics Cooling (CPU/GPU cold plate, Focused cooling, Next-gen chip thermal management)
(3) Heat Exchangers (PCHE, Shell & Tube, Plate HX, Cryogenic applications)
(4) Heat Pump Systems (Chemisorption, Adsorption, High-temperature (300°C))
(5) Low GWP Refrigerant Systems (Eco-friendly refrigerants, VLE measurement, Equation of state development)
(6) Two-Phase Heat Transfer (Boiling, Evaporation, Pulsating heat pipes, Vapor chambers)
(7) Cryogenic Heat Transfer (Liquid hydrogen vaporizer, Supercritical fluid, Sub-200°C systems)

## PUBLICATIONS

### SCI/SCIE Journal Papers

(1) **W. Kim**, M. Kim, B. Kim, K.H. Lee, T. Kim, "Advanced modeling for large-scale low-temperature PCHE design with the effect of scale-up mechanism", *Applied Thermal Engineering* 309, 133887 (2026).

(2) D.H. Kim, **W. Kim**, K.H. Lee, "VLE measurements and mixture-model parametrization for R1234yf + CF3I and R1123 + CF3I", *Int. J. Refrig.* 192, 107108 (2026).

(3) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, "Falling film evaporation of ammonia on smooth and Low-Fin tube arrays", *Thermal Science and Engineering Progress* 69, 104483 (2026).

(4) B. Kim, K.H. Lee, M. Kim, S. Moon, J.W. Yoo, **W. Kim**, "Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger", *Applied Thermal Engineering* 273, 126455 (2025).

(5) D.H. Kim, **W. Kim**, J.H. Kim, H.S. Kim, K.H. Lee, "Experimental studies on the VLE of R-32/R-125 and verification of experimental setup", *Journal of Mechanical Science and Technology* 38(10), 5779-5785 (2024).

(6) H.S. Kim, D.H. Kim, J.S. Kim, **W. Kim**, Y. Kim, "A numerical study on the performance of chemisorption heat pump according to various design conditions", *Applied Thermal Engineering* 243, 122519 (2024).

(7) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, "Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures", *Applied Thermal Engineering* 247, 122986 (2024).

(8) H.S. Kim, J.H. Kim, J.S. Kim, **W. Kim**, Y. Kim, "Experimental study of a chemisorption heat pump under different operation conditions", *Applied Thermal Engineering* 240, 122274 (2024).

(9) J. Kim, C.G. Kim, H.S. Kim, **W. Kim**, D.H. Kim, "Characterization of liquid behavior in distributor of falling-film evaporator", *Physics of Fluids* 35(6), 067124 (2023).

(10) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, S.H. Yoon, "Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement", *International Communications in Heat and Mass Transfer* 149, 107157 (2023).

(11) **W. Kim** and S.J. Kim, "Fundamental issues and technical problems about pulsating heat pipes", *Journal of Heat Transfer - Transactions of the ASME* 143, 100803 (2021).

(12) **W. Kim** and S.J. Kim, "Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes", *International Journal of Heat and Mass Transfer* 149, 119251 (2020).

(13) **W. Kim** and S.J. Kim, "Effect of reentrant cavities on the thermal performance of a pulsating heat pipe", *Applied Thermal Engineering* 133, 61-69 (2018).

### KCI Journal Papers

(1) J.S. Kim, D.H. Shin, J. Shim, K.H. Lee, S. Sohn, **W. Kim**, C. Song, "A Study on the Design of an Air-to-Air Plate Heat Exchanger for 300°C High Temperature Heat Pumps", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 38(1), 13-25 (2026).

(2) **W. Kim**, B. Kim, S. Sohn, K.H. Lee, J. Kim, "Experimental Investigation on the Freezing Condition of PCHE for Cryogenic Liquid Hydrogen Vaporizer", *Journal of Hydrogen and New Energy* 35(2), 240-248 (2024).

(3) **W. Kim**, H.S. Kim, J. Kim, D.H. Kim, "An Experimental Study on The Thermal Performance Measurement and Flow Visualization of Falling Film Evaporator Using R-1233ZD(E) Refrigerant", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 36(1), 9-17 (2024).

(4) D.H. Kim, H.S. Kim, J. Kim, **W. Kim**, "An Experimental Study on the Performance Characteristics of a Falling Film Evaporator for R-1234ze(E) Refrigerant", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 35(7), 363-370 (2023).

(5) J.S. Kim, S.A. Kim, D.H. Shin, **W. Kim**, S. Moon, Y. Chung, S. Sohn, "Comparative Study of Air Cooling and Immersion Cooling for the Thermal Management of a Cylindrical Battery Pack", *Transactions of the KSME B* 47(10), 543-550 (2023).

(6) S. Sohn and **W. Kim**, "A Study on Anti-Icing Design by Conjugate Heat Transfer Analysis in a Lab-Scale PCHE for Supply of Cryogenic High Pressure Liquid Hydrogen", *Transactions of the Korean Hydrogen and New Energy Society* 33(5), 541-549 (2022).

(7) H.S. Kim, **W. Kim**, K.H. Lee, D.H. Kim, "Numerical Study on the Performance of Hybrid Falling Film Evaporator", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 34(8), 371-379 (2022).

(8) K.H. Lee, D.H. Kim, **W. Kim**, J.S. Kim, Y. Kim, J.W. Yoo, "Selection of the Equation of State for Thermodynamic Properties in the Low GWP Mixture Refrigerants", *Transactions of the KSME B* 46(10),

563-571 (2022).

### International Conference Proceedings

(1) 2024, "Experimental and theoretical investigation on avoiding freezing phenomena of PCHE for cryogenic liquid hydrogen vaporizer", 29th ICEC

(2) 2024, "Understanding the thermal characteristics of falling film evaporation of R-1233ZD(E) refrigerant using flow visualization and heat transfer measurement", 11th ACRA

(3) 2023, "Experimental investigation on the flow and thermal characteristics of falling film evaporator using R-1233ZD(E) refrigerant", 11th ICBCHT

(4) 2019, "Experimental Investigation on the Thermal Performance of PHP Operating in a Circulation/oscillation Mode", Korea-China-Japan Joint Symposium

(5) 2019, "Comparison of thermal performance between CEPHP and CLPHP", 4th TFEC

(6) 2018, "Experimental investigation on the thermal performance of double-condenser pulsating heat pipes", 10th ICBCHT

(7) 2017, "Experimental Investigation on the Thermal Performance of a Pulsating Heat Pipe with Artificial Cavities", 1st ACTS

(8) 2017, "Experimental investigation of artificial cavities on the thermal performance of a pulsating heat pipe", InterPACK 2017

### Domestic Conference Presentations

(1) 2018, "Development of deployable radiator for spacecraft using a pulsating heat pipe", KSME 2018 Spring Annual Meeting (Thermal Engineering Division)

(2) 2018, "Effects of circulating flow on the thermal performance of a pulsating heat pipe", KSME 2018 Annual Meeting

(3) 2016, "Effects of artificial cavities on the startup performance of a pulsating heat pipe", KSME 2016 Annual Meeting

## PATENTS

*Domestic Patents (26)*

(1) Adsorption heat pump with pulsating heat pipe **[2022.12.16]**

(2) Fire-suppression cooling system for data centers **[2024.12.26]**

(3) Immersion cooling system for data centers **[2024.12.26]**

(4) Multi-thermal management device and method for immersion cooling tanks **[2024.11.14]**

(5) Tray for falling-film evaporator and falling-film evaporator **[2023.12.12]**

(6) Multi-stage evaporator **[2024.11.06]**

(7) Tray structure for heat exchanger (2) **[2022.10.13]**

(8) Tray structure for heat exchanger (3) **[2022.10.13]**

(9) Heat exchanger with anti-freezing capability (1) **[2023.03.27]**

(10) Heat exchanger with anti-freezing capability (2) **[2023.03.27]**

(11) Modular fin-tube reactor **[2024.11.04]**

(12) Fin-tube type reactor **[2024.11.04]**

(13) Tray structure for heat exchanger (1) **[2022.10.13]**

(14) Oil recovery system for refrigeration systems (1) **[2022.10.13]**

(15) Oil recovery system for refrigeration systems (2) **[2022.10.13]**

(16) Micro-channel reactor **[2023.04.19]**

(17) Immersion cooling device **[2024.10.25]**

(18) Geothermal heat supply device and heating system **[2023.10.17]**

(19) Heat exchanger thermal performance testing apparatus **[2023.06.16]**

(20) Ternary refrigerant composition and heat pump system **[2022.10.24]**

(21) Display device **[2023.10.20]**

(22) Duct assembly and combustor **[2022.03.30]**

(23) Immersion cooling HVAC system and method **[2022.12.22]**

(24) Heat pipe integrated reactor for adsorption heat pump **[2022.04.06]**

(25) Adsorption heat pump evaporator and system **[2023.03.31]**

(26) Plate pulsating heat spreader with artificial cavities (KR 10-1832432) **[2018.02.20]**

*International Patents (1)*

(1) Plate pulsating heat spreader with artificial cavities (US 10,436,520 B2) **[2019.10.08]**

## SOFTWARE REGISTRATIONS

(1) AI-based DLC cold plate automated design program **(C-2026-059148)** [2026.11.12]

(2) PCHE thickness design program **(C-2024-052179)** [2024.12.11]

(3) High-temperature heat pump cycle design program **(C-2024-038691)** [2024.10.22]

(4) Heat exchanger HTC uncertainty analysis program **(C-2023-059071)** [2023.12.12]

(5) Lab-scale heat exchanger analysis program **(C-2023-056463)** [2023.11.30]

(6) 100 kg/hr hydrogen PCHE design program **(C-2025-056891)** [2025.12.11]

(7) Adsorption cooling simulation program **(C-2025-034652)** [2025.09.03]

(8) Vapor chamber performance analysis program **(C-2024-041200)** [2024.11.04]

(9) Fin-tube heat exchanger design program **(C-2024-038692)** [2024.10.22]

## TECHNOLOGY TRANSFERS

(1) Phase-equilibrium property evaluation for refrigerant mixtures

(2) Heat transfer enhancement for data center liquid cooling

(3) Low-temperature freezing performance test of a PCHE for vehicle heat exchangers

(4) Diffusion bonding technology for cryogenic hydrogen heat exchangers

(5) Ultra-thin vapor chamber design technology

(6) Heat pump design for low-dew-point dehumidifiers in battery manufacturing

(7) Compact inverter compressor-based dress care system

## RESEARCH PROJECTS

(1) **Channel-Scale Transition Behavior and Visualization-Based Design Guidelines for Low-GWP Refrigerant Two-Phase Direct-to-Chip Cold Plates** (PI) [2026-2028]

(2) **Development of an AI-based Automated Design Tool for Data Center Direct Liquid Cooling (DLC) Systems** (PI) [2026]

(3) **Development of a Direct Liquid Cooling (DLC) System for Reducing Power Consumption in Data Centers** (Lead) [2026-2029]

(4) **Jet-Enhanced Immersion Cooling for Data Centers** (PI) [2025]

(5) **KIMMPROP - Thermophysical Property iOS/Android App** (PI) [2025]

(6) **PCHE for Liquid Hydrogen Supply System** (Lead) [2021-2025]

(7) **Compact PCHE Design for Liquid Hydrogen** (Lead) [2022-2026]

(8) **Immersion Cooling Waste Heat Utilization** (Lead) [2024-2028]

(9) **Development of Eco-Friendly, High-Efficiency, Non-Toxic Refrigerants using AI and Heat Pump Application and Verification Technologies** (Participant) [2026-2029]

(10) **300°C High-Temperature Heat Pump System** (Participant) [2023-2028]

(11) **Chemisorption Heat Pump with Electrochemical Compressor** (Participant) [2021-2025]

(12) **Next-Gen Alternative Refrigerant & HVAC Technology** (Participant) [2021-2025]

(13) **Low-Charge Industrial Heat Pump** (Participant) [2022-2023]

(14) **Water-Source Heating/Cooling Hybrid System** (Participant) [2022-2023]

(15) **Heat Pipe HX Performance Enhancement** (Participant) [2022]

(16) **Post H-class Gas Turbine Technology** (Participant) [2022]

(17) **Compact Inverter Compressor Dress Care System** (Participant) [2022]

(18) **Ultra-Thin Vapor Chamber Performance Enhancement** (Participant) [2022-2023]

(19) **PCHE Freezing Test for Vehicle Heat Exchangers** (PI) [2025]

(20) **BTMS Integrated A/Con System Evaluation** (Participant) [2025]

(21) **3D Multiporous Cooling System for Ultra-high Heat Flux Applications** (Graduate Researcher) [2021]

(22) **Smart protective suit with active cooling/ventilation and antivirus suit changing system** (Graduate Researcher) [2020]

(23) **Next-Generation CPI Reliability Structural & Thermal Design** (Graduate Researcher) [2020]

(24) **OHP High-Pressure Working-Fluid Charging Technology** (Graduate Researcher) [2020]

(25) **Thermal optimization of MMU rear housing for reducing the weight** (Graduate Researcher) [2019-2020]

(26) **Development of a Flexible Thin-Film Ultra-Thermal-Conductor** (Graduate Researcher) [2015-2020]

(27) **Study on a smart space thermal management system** (Graduate Researcher) [2016-2018]

## PROFESSIONAL ACTIVITIES

(1) **Council Member**, Korean Society of Mechanical Engineers (KSME) [2026.04-Present]

(2) **Committee Member**, KSME Thermal Engineering Division, Future Development Committee

(3) **Committee Member**, SAREK Future Growth Special Committee

(4) **Committee Member - Thermal Management / Cooling**, SAREK Data Center Facility Technology Division Committee [2026.03-Present]

(5) **Collaborating Researcher / Mentor**, 2026 Artience Daejeon (Daejeon Cultural Foundation art-science program) [2026.03-2026.12]